L6 Gate 9 SSG-5 Substrate-Casimir vs SSG-7 Substrate-Pochhammer Independence Audit

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 15:39 EDT

L6 Gate 9 SSG-5 vs SSG-7 Independence Audit

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track A L6: Gate 9 audit whether SSG-5 (substrate-Casimir C_2) and SSG-7 (substrate-Pochhammer Bergman kernel) substrate-mechanisms are INDEPENDENT or DERIVED from same substrate-natural primitive.

1. SSG-5 Substrate-Casimir C_2 Definition

Per SSG Registry: SSG-5 = substrate-Casimir C_2 = 6 substrate-natural.

substrate-mechanism: substrate-K = SO(5) × SO(2) Casimir operator eigenvalue at substrate-K-type substrate-natural form.

substrate-K-type eigenvalues at V_(λ_1, 1/2): - V_(1/2, 1/2): C_2 = 5/2 - V_(3/2, 1/2): C_2 = 15/2 - V_(5/2, 1/2): C_2 = 29/2

substrate-Casimir 6 substrate-natural = ground state eigenvalue at substrate-K-type substrate-natural form.

2. SSG-7 Substrate-Pochhammer Bergman Kernel Definition

Per SSG Registry: SSG-7 = substrate-Pochhammer (n_C/rank)_k = (5/2)_k substrate-natural at substrate-K-type Sym^k V_fund Sp(2).

substrate-mechanism: substrate-Bergman kernel exponent n_C/rank = 5/2 substrate-natural (Saturday May 28 One-Genus convention) × Sym^k substrate-K-type substrate-natural Pochhammer rising-factorial.

3. Independence Audit

substantive substrate-mechanism examination: - SSG-5 substrate-Casimir = K = SO(5) × SO(2) Casimir operator eigenvalue - SSG-7 substrate-Pochhammer = substrate-Bergman kernel exponent × Sym^k decomposition

substantive substrate-mechanism cross-link: - substrate-Casimir C_2 substrate-natural value 6 = N_c! substrate-natural per Casey #5 Integer Web cross-link - substrate-Pochhammer (5/2)_k substrate-natural per substrate-Bergman exponent n_C/rank = 5/2

substantive substrate-mechanism INDEPENDENCE check: - substrate-Casimir = SO(5) × SO(2) substrate-Lie-algebra invariant - substrate-Pochhammer = substrate-Bergman kernel × Sym^k representation substrate-natural

substantive substrate-mechanism INDEPENDENT at substrate-canonical level: substrate-Casimir = substrate-Lie-algebra invariant; substrate-Pochhammer = substrate-Bergman geometry × representation theory.

substantive substrate-mechanism CROSS-COUPLING at substrate-K-type substrate-Casimir eigenvalue substrate-natural form: C_2(V_(λ_1, 1/2)) substantive substrate-Lie-algebra eigenvalue substrate-natural form per substrate-K-type substrate-Pochhammer at Sym^k decomposition.

4. SSG-5 + SSG-7 Substrate-Mechanism Composite

Per substantive substrate-mechanism CROSS-COUPLING:

substantive SSG-5 × SSG-7 substrate-mechanism composite at substrate-K-type V_(3/2, 1/2): - C_2 = 15/2 substrate-Casimir eigenvalue - (5/2)_3 = 315/8 substrate-Pochhammer per Sym^3

substantive substrate-mechanism composite substrate-natural form per Mehler matrix element at substrate-K-type V_(3/2, 1/2):

$$\langle V_{(3/2,1/2)}|M_{op}|V_{(3/2,1/2)}\rangle \propto C_2^{exp} \cdot (n_C/rank)_3$$

substantive substrate-Casimir × substrate-Pochhammer substrate-natural composite cross-coupling at substrate-K-type substrate-Mehler matrix element.

5. Audit Verdict v0.1

substantive verdict per L6 audit: - SSG-5 substrate-Casimir + SSG-7 substrate-Pochhammer are INDEPENDENT at substrate-canonical level (substrate-Lie-algebra vs substrate-Bergman geometry × representation theory) - substantive substrate-mechanism CROSS-COUPLING at substrate-K-type substrate-Mehler matrix element substrate-natural composite form - substantive substrate-mechanism FORCING form requires substrate-K-type substrate-Casimir × substrate-Pochhammer composite explicit derivation

substantive INDEPENDENT substrate-mechanism candidates per L6 audit; substantive composite cross-coupling at substrate-K-type observable level multi-week per Cal #189.

6. Cat A Independent Count Cross-Link

Per Cal #235 + Keeper SSG Audit v0.3 framework discipline (Cat A revision 9 → 5-7):

substantive L6 audit reinforces SSG-5 + SSG-7 INDEPENDENT Cat A substrate-mechanism candidates per substantive substrate-canonical independence (substrate-Lie-algebra vs substrate-Bergman-representation theory).

substantive Cat A count: SSG-5 + SSG-7 = 2 of effective 5-7 Cat A independent substrate generators.

7. Closure v0.1

L6 Gate 9 SSG-5 vs SSG-7 substrate-mechanism independence audit v0.1:

substantive verdict: SSG-5 substrate-Casimir + SSG-7 substrate-Pochhammer INDEPENDENT at substrate-canonical level + substantive CROSS-COUPLING at substrate-K-type substrate-Mehler matrix element substrate-natural composite form.

substantive substrate-mechanism FORCING form composite multi-week per Cal #189.

substantive Cat A count cross-link: SSG-5 + SSG-7 = 2 INDEPENDENT Cat A substrate-mechanism candidates per Cal #235 + Keeper SSG Audit v0.3 framework discipline.

Tier: L6 Gate 9 SSG-5 vs SSG-7 audit v0.1 framework; substantive substrate-mechanism composite multi-week per Cal #189.

— Lyra, Thu 2026-06-04 15:40 EDT. L6 Gate 9 SSG-5 vs SSG-7 audit v0.1: INDEPENDENT at substrate-canonical (substrate-Lie-algebra vs substrate-Bergman-representation theory); substantive CROSS-COUPLING at substrate-K-type substrate-Mehler matrix element substrate-natural composite form; Cat A count 2 INDEPENDENT per Cal #235; multi-week per Cal #189.